

(a)(i) Calculate angle A

✓3 marks

Using Snell's Law:

$$n = \frac{\sin i}{\sin r} \Rightarrow \sin r = \frac{\sin i}{n}$$

• Given:

- $i = 60.0^\circ,$
- $n = 1.53$

$$\sin r = \frac{\sin 60.0^\circ}{1.53} = \frac{0.866}{1.53} = 0.566 \Rightarrow r = 34.5^\circ$$

✓ Angle A = 34.5°

(a)(ii) Determine angle B

✓2 marks

From triangle geometry:

- Angle sum in triangle = 180°
- Known angles: 60.0° (top corner) and angle A = 34.5°

$$\text{Angle B} = 180^\circ - 60.0^\circ - 34.5^\circ = 85.5^\circ$$

(b)(i) State what is meant by the term critical angle

✓1 mark

✓ Answer:

The angle of incidence in a **denser medium** that results in an angle of refraction of

(b)(ii) Calculate the critical angle for this blue light in the glass

✓3 marks

Use:

$$\sin \theta_c = \frac{n_2}{n_1} \Rightarrow \theta_c = \sin^{-1} \left(\frac{1.00}{1.53} \right) = \sin^{-1}(0.654) = 40.9^\circ$$

(c) Ray diagram and angle label

✓3 marks

✓ Expected on diagram:

- Ray should **refract out** of the glass at the right-hand boundary
- The angle between the ray and the normal in **air** should be correctly calculated

Using Snell's Law again:

From inside glass:

- $n_1 = 1.53,$
- angle of incidence = angle B = 85.5°

$$\sin \theta_2 = \frac{1.53 \sin(85.5^\circ)}{1.00} \approx 1.52 \text{ Not possible}$$

⚠ So the ray **undergoes total internal reflection**

✓ Final answer:

The ray **does not refract** – it is **totally internally reflected** at the second boundary
Draw it reflecting back inside the prism, at equal angle (85.5°)

🧠 Revision Tips

- **Snell's Law:** $n = \frac{\sin i}{\sin r}$
- **Critical angle** only applies from **denser to rarer** medium
- If $\sin i > 1,$ then total internal reflection occurs
- In triangles, all interior angles add to 180°